



Comparative Study of One-Handed vs Two-Handed EEG Intent Recognition for Applications in Human-Robot Interaction

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ABSTRACT

Brain-computer interfaces (BCI) act as a useful tool, allowing subjects with disabilities to regain control over their bodily functions. These interfaces can classify signals from the brain using machine learning and applied to human-computer or human-robot interaction. The motor movements that the subject executes to operate the BCI may vary, and it is important to assess the execution of different muscle groups and movements for accurate performance and user feedback. Existing studies focus on single-arm, or unilateral, movements. Currently, research is alternating to include bimanual movements. In this paper, we assess the efficacy of various motor movements within a BCI by evaluating the classification of unimanual and bimanual motor intent, alongside the incorporation of sequential alternating unilateral movements, employing Logistic Regression (LR) and Common Spatial Patterns (CSP) methodologies. Four scenarios in which the user is prompted to move designated arms either left or right: 1) unilateral movement of the user's dominant arm, 2) unilateral movement of the user's nondominant arm, 3) sequential alternating unilateral movements of both arms in their respective directions, and 4) mirrored bimanual movement of the arms. This study was based on data acquired from 11 healthy, right-handed. The results show coordinating sequential alternating unilateral movements outperforms single unimanual and bimanual classification for the BCI pipeline with an accuracy of $81.5\% \pm 7.7\%$ using EEG signals for logistic regression. Thus, it is recommended to prioritize sequential alternating unilateral movements during the

training of BCI-based human-robot interaction to achieve optimal performance.

CCS CONCEPTS

• Human-computer interaction; • Machine learning; • Emerging technologies;

KEYWORDS

Brain-Computer Interaction (BCI), Human-Robot Interaction (HRI), Rehabilitation, Electroencephalography (EEG) classification

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1 INTRODUCTION

BCIs are systems that facilitate users' ability to interact with their environment or devices by acquiring neurophysiological signals from the brain. BCI development is driven by its potential to aid those with losses in sensory or motor functions, such as locked-in patients, stroke survivors or those with spinal cord injuries, in regaining control over their bodies [1]. BCIs have been used to assist stroke patients speak again by tracking brain activity related to the vocal tract and even facilitating movement restoration [1, 2]. The application of BCIs extends from rehabilitation [3, 4] and diagnosis to gaming, virtual reality, and human-robot interaction [5]. Research has also explored the potential of BCIs in high throughput image searches, leveraging the human brain's proficient capacity for visual processing, as well as their cognitive enhancement ability through monitoring alertness [4], cognitive load, and other uses in education and learning.

BCIs typically operate through three main processes: signal acquisition and processing, pattern recognition via machine learning,

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and sensory feedback. For signal acquisition, BCIs must read neurophysiologic measurements from or around the brain, which can be achieved through various techniques. These techniques can be broadly divided into invasive and noninvasive methods. Noninvasive methods include electroencephalograph (EEG), functional Magnetic Resonance Imaging (fMRI), functional Near Infrared (fNIR) Imaging, Positron Emission Tomography (PET), and Magnetoencephalography (MEG) [6]. This research focuses on EEG-based BCIs which consist of electrodes placed on specific areas of the surface of the user's scalp to monitor the electrical potential of a large population of neural activity. The number of electrodes can vary, and they are designated for specific brain areas [7].

The acquired signals undergo preprocessing, feature extraction, and classification. Processing techniques include frequency and spatial filtering. Common filters include Finite Impulse Response Filters (FIR), Infinite Impulse Response (IIR) [8], Common Average Reference, Independent Component Analysis, Principal Component Analysis (PCA), and Common Spatial Filtering. Some researchers have used a bandpass FIR filter was implemented to reduce noise within the signals and extract specific frequencies from BCI electrodes [9]. The frequencies of interest found in the sensorimotor cortex area range 6-30 Hz. This range correlates with the μ (7.5-12.5 Hz) and β (12.5-30 Hz) brain waves, which heavily indicate motor execution. In this paper, a bandpass FIR filter was implemented to reduce noise within the signals and extract specific frequencies from BCI electrodes [9]. The frequencies of interest found in the sensorimotor cortex area range 6-30 Hz. This range correlates with the μ (7.5-12.5 Hz) and β (12.5-30 Hz) brain waves, which heavily indicate motor execution.

Common Spatial Patterns (CSP) is a supervised spatial filter commonly used for feature extraction in motor imagery [10]. This filter finds a projection that maximizes the variance of the components of an average event-related, allowing for the discrimination between two conditions. Using this method requires using the variance in a classification setup. Machine learning algorithms are used for classification including linear classifiers such as Linear Discriminant Analysis (LDA), Support Vector Machine (SVM), and Logistic Regression (LR). Deep Neural networks like Convolutional Neural Networks and Recursive Neural Networks are also commonly used [11, 12]. Previous work has been done to access the performance of these algorithms in different training setups, such as online and offline training [20].

A fair portion of BCI research focuses on the rehabilitation of single muscle groups working independently rather than bilateral muscle groups working together. Everyday tasks tend to require multiple muscle groups to work in synchronous harmony rather than a single muscle group. Additionally, focusing solely on a single muscle group may enable a disabled patient to grasp or move a glass but would be inadequate for more complex tasks, such as typing or maintaining balance while running. Previous research [13] suggests that bilateral actions result in improved motion prediction when monitoring the .3 -3 Hz frequency range. A review of several studies focusing on bimanual motion [14] shows that involving both left and right armed movements should be easier to detect due to spreading the neural activity to different regions of the brain.

While most studies focus on unilateral movements of a single arm, there is a rising number of papers focused on bimanual movements due to their resemblance to real-world applications [15, 16]. These papers compare the performance of unimanual and bimanual motor movements based on signals from the brain. In this paper, the performance of the CSP-LR pipeline was studied to compare sequential alternating unilateral movements in addition to conventional unilateral and bilateral movements, focusing on assessing their potential for robotic control. Sequential alternating unilateral movements, or coordinated unilateral movements for brevity, serve as a variation to conventional unilateral and bimanual movements where users are switching between control of each arm. 11 healthy subjects participated in the experiments. Each subject performed four separate motor movements for left and right binary classification to test how these movements contribute to the accuracy of the predicted output and performance of the system [13]. These four movements were selected due to their ease of learnability, comparable to movements in similar studies [15, 16], and structured around the subject sitting at the resting state with elbows supported, and hands resting on a table. The movements consisted of four options:

- Unimanual lateral rotation of the right arm away from the body's midline during the right directional prompt; medial rotation of the right arm towards the body's midline during the left directional prompt, as shown in Fig. 1a
- Unimanual medial rotation of the left arm towards the body's midline during the right directional prompt; lateral rotation away from the body's midline for the left directional prompt, as shown in Fig. 1b
- Alternating sequential unimanual lateral rotation of the right arm away from the body's midline during the right directional prompt, and lateral rotation of the left arm away from the body's midline for the left directional prompt, as shown in Fig. 1c
- Mirrored bimanual medial rotation of left arm towards the body's midline and lateral rotation of the right arm away from the body's midline during the right directional prompt, lateral rotation of the left arm away from the body's midline and medial rotation of the right arm towards the body's midline for the left directional prompt., as shown in Fig. 1d

The main contribution of this work is the inclusion of sequential alternating unilateral movements in the use of a BCI in addition to conventional unilateral and bilateral movements. Experimental results show that the pipeline predicted the output of the EEG data better using a motion that requires both right and left arms in a coordinated fashion (option 3), suggesting that this input movement should be exercised in rehabilitation tasks typically employed during robotic rehabilitation exercises.

2 MATERIALS AND METHODS

This section will describe the materials used for data acquisition, preprocessing, and feature extraction and classification.



Figure 1: Four movements options.

2.1 EEG Hardware and Data

OpenBCI [17] is an open-source platform for neuroscience and brain-computer interfacing. It provides a range of tools for biosensing and data acquisition relevant to BCIs. The platform is used for this project. The first tool used is the EEG

Electrode Cap and Cyton + Daisy Biosensing Board. Electrodes on the EEG cap are chosen to focus on motor execution. Of the 16 available on the Cyton+ Daisy board, 15 channels were used. The frontal pole (FP1-FP2), frontal lobe (F3- F4, F7-F8, Fz), central region (C3, Cz, C4), parietal (P3, Pz, P4) and occipital (O1, O2) electrode locations were selected, as shown in Figure 2.

Once set up the OpenBCI_GUI is accessed, featured in Figure 3, where the raw data is collected. From here, data can be streamed via the Networking menu through the Lab System Layer (LSL) from any data format.

2.2 Experimental Design and Setup

To begin, the experiment was done on 11 subjects consisting of 5 males and 6 females ranging from age 19 to 33 and an average age of 23. Younger adult subjects were utilized to minimize the

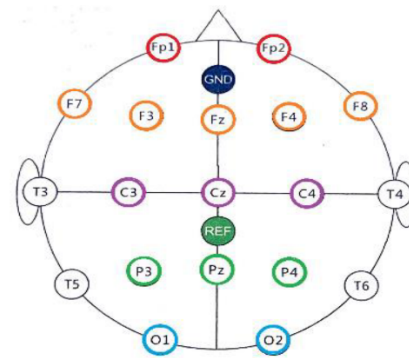


Figure 2: Electrode placement

impact of additional SMR noise which may be present in older adults. Subjects were informed about the experimental procedure and operation of the BCI after being asked to fill out a consent form and an Edinburgh Handedness Inventory (EHI) [18] to determine the laterality index (LI) of each subject. Scores closer to 100 indicate



Figure 3: OpenBCI GUI showing time series data, FFT plot, Head Plot, and Focus Widget.

Table 1: Subject Information

Subject	Sex	LI
1	F	90
2	F	100
3	M	100
4	F	90
5	M	80
6	M	80
7	M	90
8	M	80
9	F	90
10	F	100
11	F	90
Mean $\pm$ Std	0.737 $\pm$ 0.026	90.9 $\pm$ 7

the subject's dominant hand is the right, while closer to 0 indicates the left arm as the dominant. All 11 subjects scored strongly in the right LI as seen in Table 1. The EEG cap is put on the subject. Using the OpenBCI GUI, the signal integrity is validated to ensure the signal strength of each channel and that there is no packet loss.

2.3 Calibration Data Acquisition

Labeled data is required to train the CSP and LR model. To create this data a lab recorder is used to take in the LSL stream from the OpenBCI GUI [17] and a marker stream. These separate streams are overlaid into one stream of labeled time series data. A Python script is used to prompt the user to perform the specific tasks and a label of "0" or "1" is sent as a stream marker. An example of the prompt can be seen below in Figure 4 prompting the user to move in the right direction. The same prompt was used for all subjects and tests to minimize the premovement processing time.

This procedure is done for each of the 4 movement types by collecting 60 trials of each class, resulting in 120 trials for each run. Two runs are done to create training and testing data for each movement set.

2.4 Pipeline Design

The pipeline employs FIR Filter, Common Spatial Patterns, and Logistic Regression, as illustrated in Figure 5. The LSL Input, Dejitter Timestamps, and Inject Calibration Data nodes perform the task of data acquisition and calibration (training) of the pipeline. This setup collects data and inserts a marker to begin and end the calibration period. The data is sent to the Inject Calibration Data node,



Figure 4: Motor Action Prompt for Movement 2.

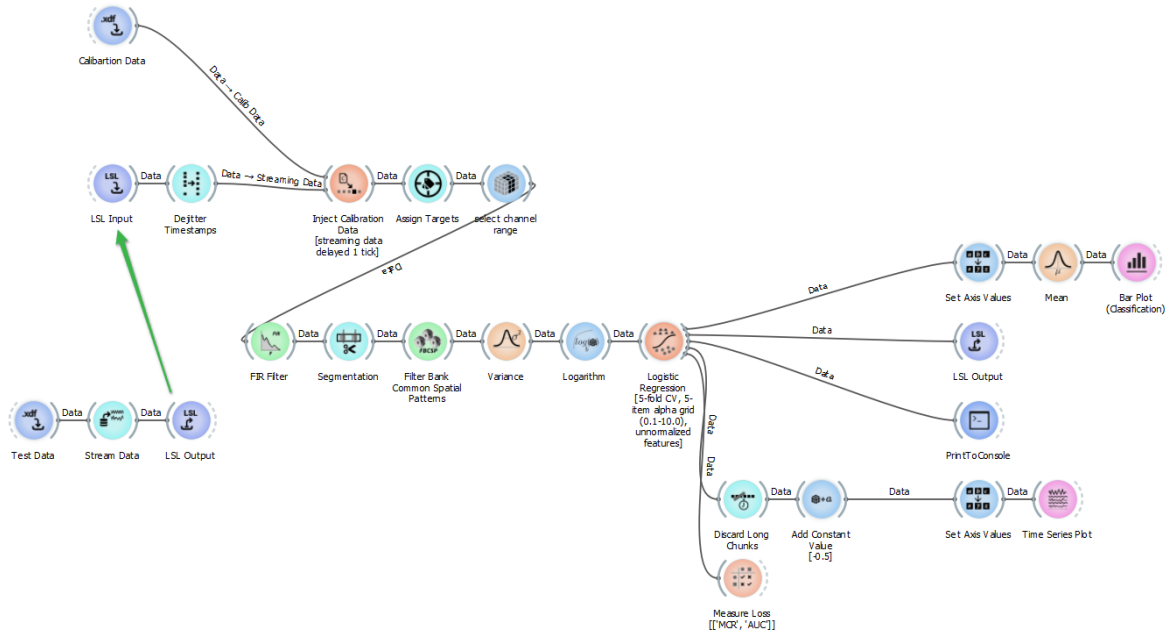


Figure 5: NeuroPyPe Pipeline Implementation of Offline Calibration, FIR Filter, CSP, and LR

which is used to pass the initial calibration data into the pipeline before the actual data is processed.

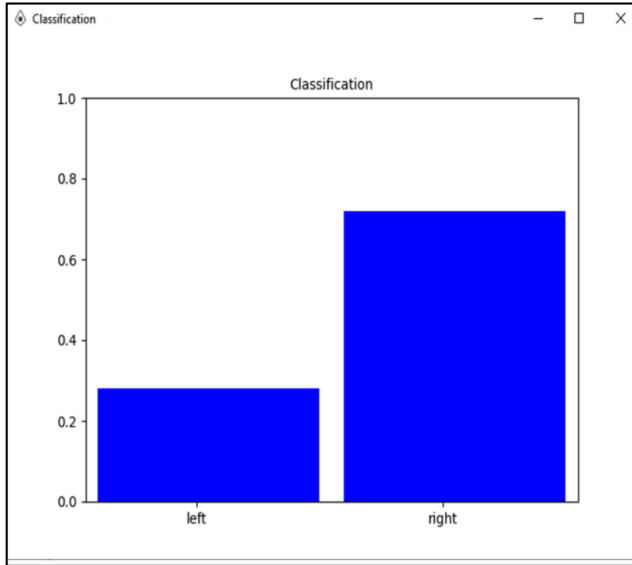
The pipeline is constructed to preprocess the data with the Assign Targets, Select Range, FIR filter, and Segmentation nodes. The Assign Targets node creates integer values associated with each marker label, while the FIR filter restricts the frequency from 7 to 30 Hz, removing undesired frequencies that will create noise. The Segmentation node then partitions the signal data according to the timing of the markers, creating segments 0.5 seconds to 3.5 seconds after the marker. Once the data passes into the Spectral Selection node, used to again filter the streams down to 7-30 Hz, the pipelines

perform CSP, then take the Log of the variance of these outputs via the Variance and Logarithm nodes before entering the LR node to perform classification.

The pipeline is set to display the predictions via a bar plot, as shown in Figure 6, where it will adjust based on the current prediction of the binary classifier. This prediction can also be seen via the output of an associated Python script and the NeuroPyPe [19] terminal.

Table 2: Performance Results

Subject	Movement 1:Right Unimanual		Movement 2:Left Unimanual		Movement 3:Coordinated Unimanual		Movement 4:Mirrored Bimanual	
	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
1	0.75	0.815	0.75	0.821	0.8	0.843	0.718	0.808
2	0.708	0.789	0.733	0.819	0.783	0.839	0.684	0.772
3	0.758	0.862	0.733	0.801	0.742	0.794	0.692	0.735
4	0.733	0.823	0.808	0.887	0.825	0.895	0.883	0.937
5	0.742	0.817	0.958	0.992	0.675	0.745	0.658	0.738
6	0.783	0.855	0.817	0.897	0.85	0.925	0.717	0.782
7	0.683	0.749	0.667	0.736	0.875	0.956	0.75	0.828
8	0.744	0.839	0.692	0.77	0.778	0.827	0.667	0.734
9	0.725	0.773	0.758	0.822	0.792	0.855	0.75	0.86
10	0.742	0.803	0.742	0.796	0.892	0.947	0.767	0.813
11	0.744	0.812	0.692	0.749	0.957	0.983	0.752	0.817
Mean	0.737	0.812	0.759	0.826	0.815	0.873	0.731	0.802
Std	0.026	0.034	0.08	0.074	0.077	0.074	0.063	0.061

**Figure 6: Live bar plot of predicted class.**

2.5 Testing

The performance of the pipeline is determined via the MeasureLoss node, which is designed to output a variety of performance metrics. For this paper, the MeasureLoss node was configured to provide a misclassification rate (MCR) and Area Under Curve (AUC) metric for each trial. The pipeline's performance is assessed by AUC and Accuracy (ACC) where $Accuracy = 1 - MCR$, measuring how often the classifier produced a correct prediction. AUC measures the area under the Receiver Operating Characteristic (ROC) curve, a graph plotting the true positive rate versus the false positive rate. This value indicates the system's ability to distinguish between the two classes. Additional qualitative assessment includes subject

feedback reported on the ease of performing these four movement types.

3 RESULTS AND DISCUSSIONS

In Table 2 the performance of each movement study, single and dual arm, can be seen side by side in reference to ACC and AUC metrics under 4 trials for each of the with 11 subjects who participated in this study. Our experiments were conducted using approval from the Internal Review Board (IRB) number 19.0329, with a continuing review approved on 3/28/2022.

From the table above, right arm only motion resulted in a mean accuracy of $0.737 \pm .026$. This improved slightly with left arm only motion resulting in a mean accuracy of $0.759 \pm .08$. The highest mean accuracy was observed during the trial where the left arm performed the left motion, and the right arm performed the right motion, resulting in a mean accuracy of $0.815 \pm .077$. Thus, the best option observed corresponds to Motion 3 (coordinated unimanual). Furthermore, histograms have been computed to show the difference in the performance for each movement for Accuracy in Figure 7.

AUC performance for each movement is shown in Figure 8.

To assess statistical significance, repeated measures analysis of variance (ANOVA) was conducted on accuracy and AUC metrics across the four movement conditions. The extremely low p-values ($p < 0.001$) for both accuracy and AUC indicate highly significant differences between the movement conditions. Specifically, the p-value of 0.0002 for accuracy and 0.0001 for AUC provide strong evidence to reject the null hypothesis of no difference between conditions. These results demonstrate the movement type significantly impacts the BCI's ability to classify brain signals. Post-hoc pairwise tests are recommended to identify which specific movement pairs differ significantly.

However, the data suggests the coordination of sequential alternating unilateral arm movements elicited distinct, discriminable neural patterns, leading to superior BCI classification performance

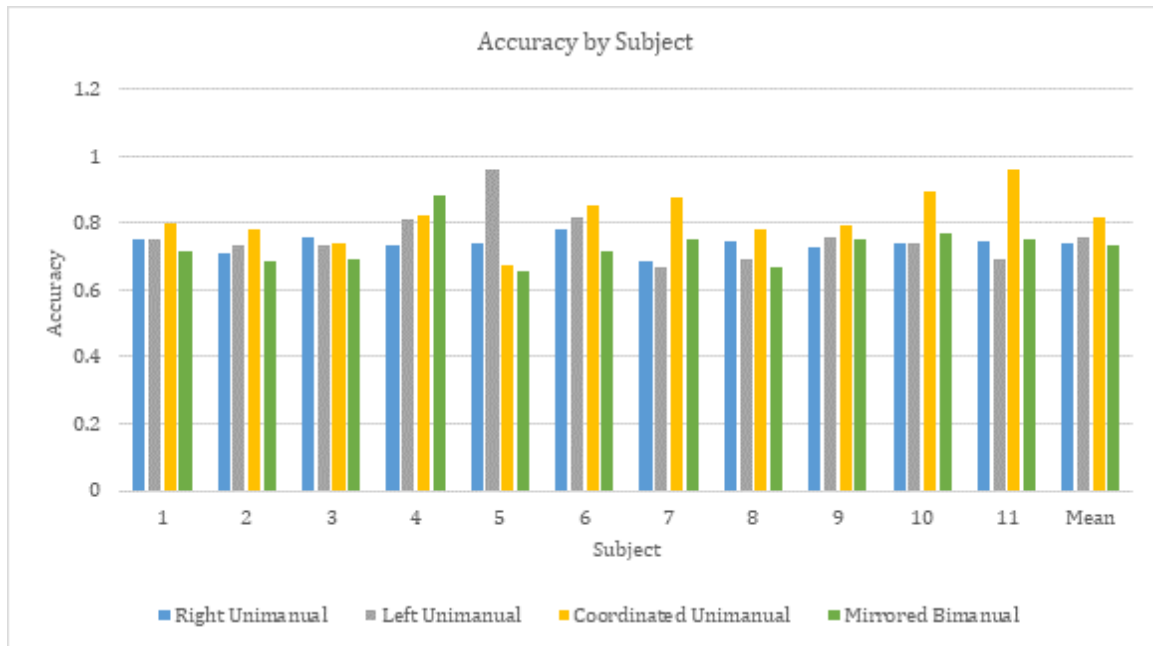


Figure 7: Bar Graph results showing the Accuracy

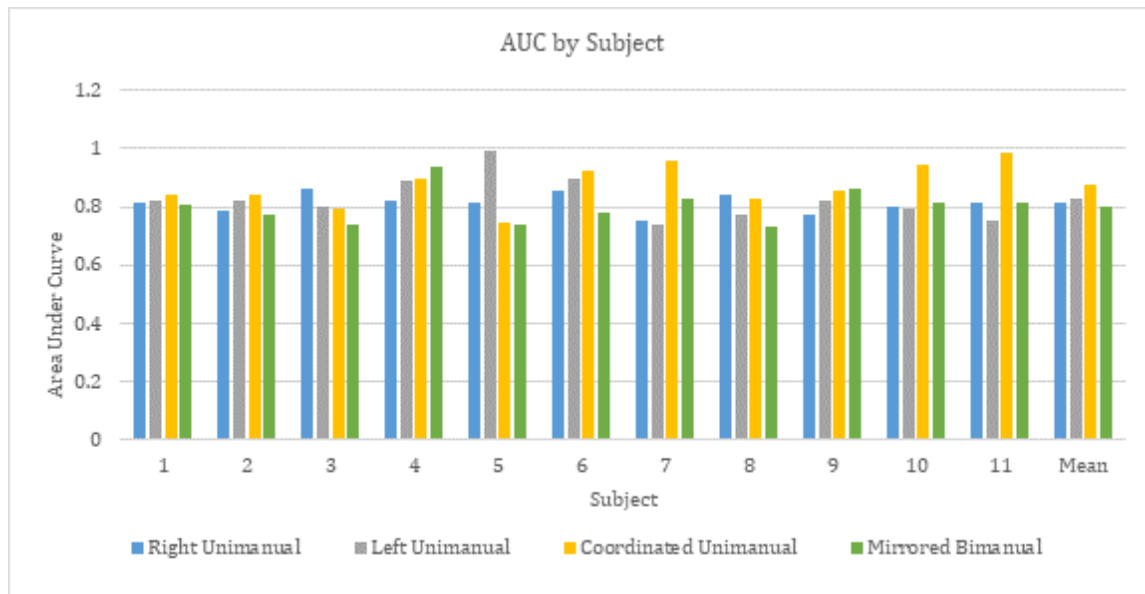


Figure 8: Graph results showing the Area Under Curve

compared to other conditions. These statistically significant findings have important implications for optimizing BCI-based rehabilitation protocols by prioritizing sequential alternating unilateral arm movements that yields the best decoding accuracy. Overall, the p-value results provide robust statistical support for the study's conclusions regarding the effect of movement type on BCI performance.

The results show that for this experiment the pipeline predicted the output of the EEG data better using a motion that requires both right and left arms. This type of motion falls in line with natural motor function, spreading the neural activity over several regions of the brain. The pipeline's prediction was hindered during motion 4, the mirrored movement removes the advantages of movement

3. By having both arms mirror each other instead of moving independently, large regions of neurons are firing making isolating specific regions related to movement more difficult. With the use of alternating sequential unilateral movements during training the user is coordinating between unilateral movements. Verbal user feedback offered by subjects during experiments also indicated that motion 3 could be more intuitive, allowing the subject to focus less on motion planning.

CONCLUSION AND FUTURE WORK

BCI-based motor execution classification is a promising area of research with the potential to revolutionize how people with disabilities interact. It is evolving into a technology beyond rehabilitation and into entertainment and human-robot interaction. There are many ways to acquire brain signals, such as EEGs and other noninvasive methods. These signals must be preprocessed and, through machine learning, can be classified into outputs for devices. Platforms such as OpenBCI and Neuropype are open-source platforms for interfacing and processing these types of signals and can be combined with methods such as FIR, CSP, and LR to predict outputs. Calibration data can be collected online to expedite the training process without the need for pre-recorded data.

The classification of different movement modalities in previous papers focuses on the assessment of conventional bimanual and unilateral movements. This paper includes sequential alternating unilateral movements when assessing the performance of a BCI motor intent classifier and current unilateral and bimanual movements. The results suggest movements used to train and operate the system have a potential role in the performance of a BCI and should be considered for applications such as robotic control and rehabilitation.

Due to limitations, such as the potential for age-related SMR noise, not represented in the subject pool, future research may seek to create more robust processes by increasing the number of subjects, subject age range, and lateral index. Further research will be conducted on a greater set of movements, as well as the implementation of a multi-class classifier to perform a hierarchical approach to detecting complex and compound movements by training parallel pipelines on the detection of simpler movements. Additionally, future research will use the results from this study to implement a robotic arm to provide users with visual feedback and to perform further testing.

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